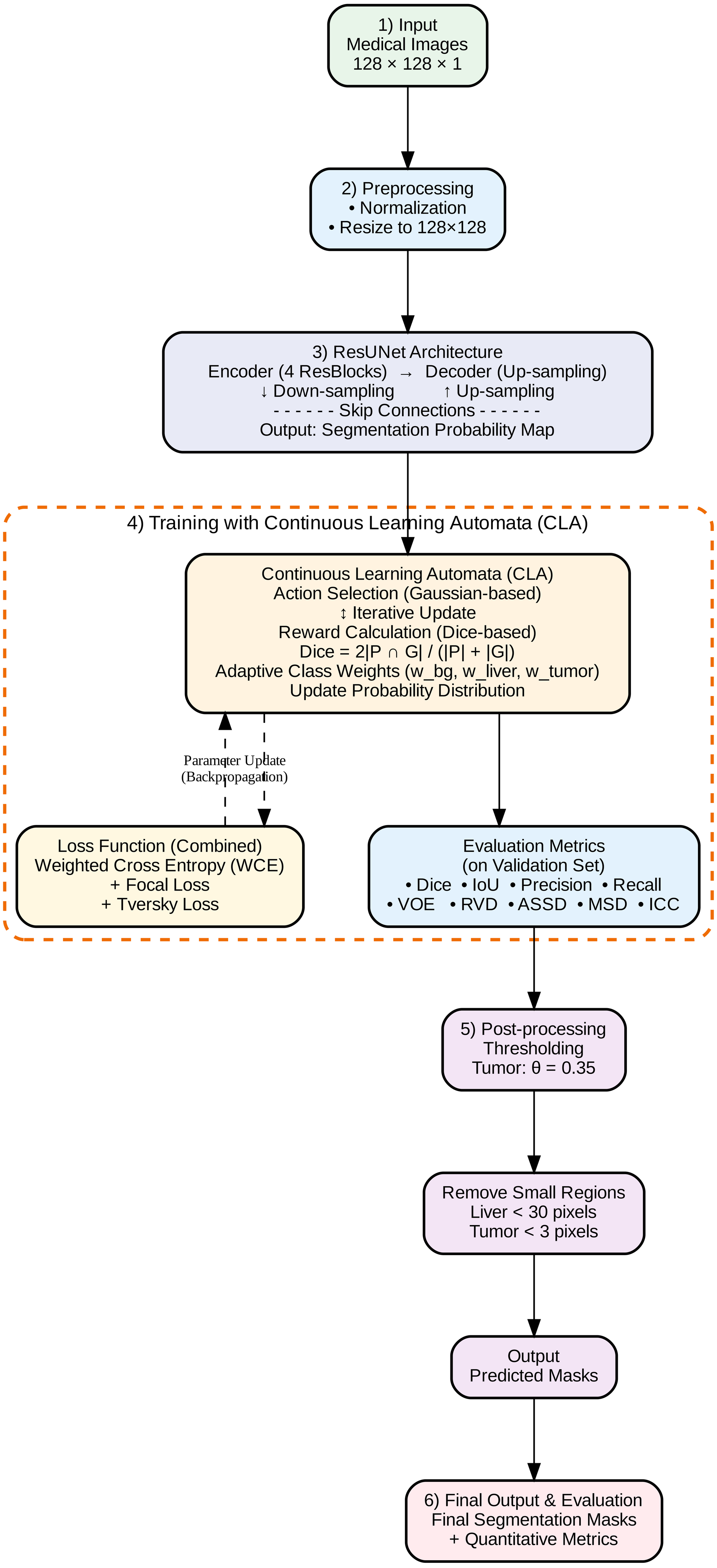
1) Input
Medical Images
128 × 128 × 1
2) Preprocessing
• Normalization
• Resize to 128×128
3) ResUNet Architecture
Encoder (4 ResBlocks) → Decoder (Up-sampling)
↓ Down-sampling ↑ Up-sampling
- - - - - - Skip Connections - - - - - -
Output: Segmentation Probability Map
4) Training with Continuous Learning Automata (CLA)
Continuous Learning Automata (CLA)
Action Selection (Gaussian-based)
↕ Iterative Update
Reward Calculation (Dice-based)
Dice = 2|P ∩ G| / (|P| + |G|)
Adaptive Class Weights (w_bg, w_liver, w_tumor)
Update Probability Distribution
Parameter Update
(Backpropagation)
Loss Function (Combined)
Weighted Cross Entropy (WCE)
+ Focal Loss
+ Tversky Loss
Evaluation Metrics
(on Validation Set)
• Dice • IoU • Precision • Recall
• VOE • RVD • ASSD • MSD • ICC
5) Post-processing
Thresholding
Tumor: θ = 0.35
Remove Small Regions
Liver < 30 pixels
Tumor < 3 pixels
Output
Predicted Masks
6) Final Output & Evaluation
Final Segmentation Masks
+ Quantitative Metrics